# SIMATS ENGINEERING

**SUBJECT NAME: INTERNET PROGRAMMING** **NAME: KANDUKURI KAVYA**

**SUBJECT CODE: CSA4316** **REG NO: 192425283**

# CLASSTEST-1

## Regular Expressions & Validation in JavaScript

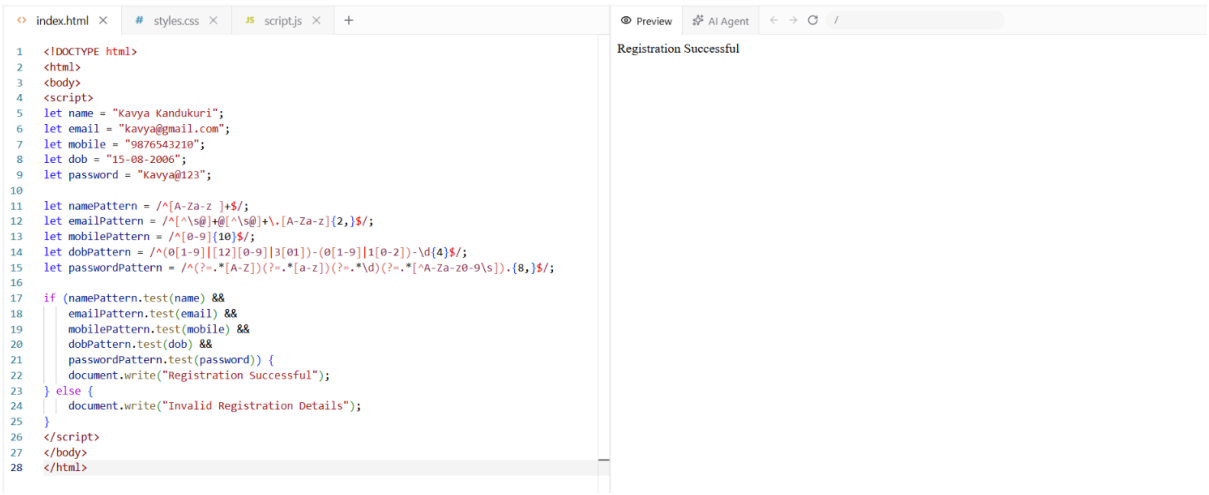
Q1: 1. Student Registration Validation: Write a JavaScript program to validate a student registration form containing a full name, email address, 10-digit mobile number, date of birth in DD-MM-YYYY format, and password. The password must contain at least one uppercase letter, one lowercase letter, one digit, and one special character, with a minimum length of eight characters.

**CODE:**

```
<!DOCTYPE html>
<html>
<body>
<script>
let name = "Kavya Kandukuri";
let email = "kavya@gmail.com";
let mobile = "9876543210";
let dob = "15-08-2006";
let password = "Kavya@123";
let namePattern = /^[A-Za-z ]+$/;
let emailPattern = /^[^\s@]+@[^\s@]+\.[A-Za-z]{2,}$/;
let mobilePattern = /^[0-9]{10}$/;
let dobPattern = /^(0[1-9]|[12][0-9]|3[01])-(0[1-9]|1[0-2])-\d{4}$/;
let passwordPattern = /^(?=.*[A-Z])(?=.*[a-z])(?=.*\d)(?=.*[^A-Za-z0-9\s]).{8,}$/;
if (namePattern.test(name) &&
    emailPattern.test(email) &&
    mobilePattern.test(mobile) &&
    dobPattern.test(dob) &&
    passwordPattern.test(password)) {
    document.write("Registration Successful");
```

```
} else {
    document.write("Invalid Registration Details");
}
</script>
</body>
</html>
```

OUTPUT:



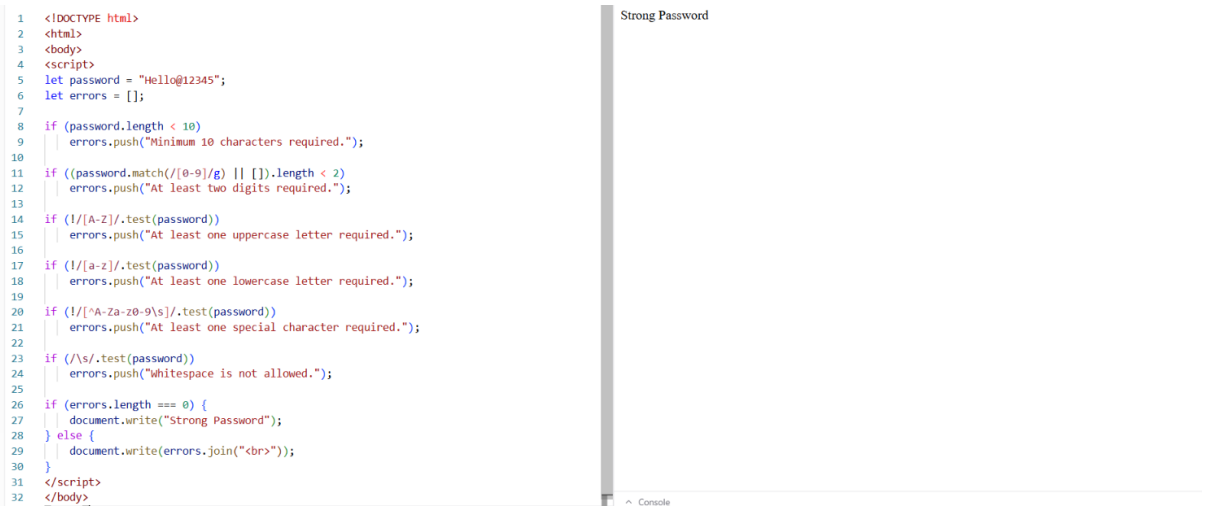

Q2: Password Strength Validation: Write a JavaScript program to validate a password with the following rules: minimum 10 characters, at least two digits, one uppercase letter, one lowercase letter, one special character, and no whitespace. Display an appropriate error message for each failed condition.

**CODE:**

```
<!DOCTYPE html>
<html>
<body>
<script>
let password = "Hello@12345";
let errors = [];
if (password.length < 10)
    errors.push("Minimum 10 characters required.");
if ((password.match(/[0-9]/g) || []).length < 2)
    errors.push("At least two digits required.");
if (!/[A-Z]/.test(password))
```

```
    errors.push("At least one uppercase letter required.");
if (!/[a-z]/.test(password))
    errors.push("At least one lowercase letter required.");
if (!/[^A-Za-z0-9\s]/.test(password))
    errors.push("At least one special character required.");
if (/\s/.test(password))
    errors.push("Whitespace is not allowed.");
if (errors.length === 0) {
    document.write("Strong Password");
} else {
    document.write(errors.join("<br>"));
}
</script>
</body>
</html>
```

**OUTPUT:**



Q3: Email Address Validation: Write a JavaScript program to validate email addresses according to these rules: the address must contain exactly one @, a non-empty username, a domain name, and a valid-looking extension of at least two letters. Reject spaces and consecutive dots.

**CODE:**

```
<!DOCTYPE html>
<html>
<body>
<script>
let email = "student@gmail.com";
let pattern = /^[^@\s.]+(?:\.[^@\s.]+)*@[^@\s.]+(?:\.[^@\s.]+)*\.[A-Za-z]{2,}$/;
if (pattern.test(email)) {
    document.write("Valid Email Address");
} else {
    document.write("Invalid Email Address");
}
</script>
</body>
</html>
```

**OUTPUT:**



Q4: Username Validation: Write a JavaScript program to validate a username that must start with a letter, contain only letters, digits, and underscores, have a length of 6–15 characters, and must not end with an underscore or contain consecutive underscores.

**CODE:**

```
<!DOCTYPE html>
<html>
<body>
<script>
let username = "Kavya_123";
let pattern = /^(?=.{6,15}$)[A-Za-z][A-Za-z0-9_]*[A-Za-z0-9]$/;
if (pattern.test(username) && !username.includes("__")) {
    document.write("Valid Username");
} else {
    document.write("Invalid Username");
}
</script>
</body>
</html>
```

**OUTPUT:**



Q5: Extract and Analyse Data: Write a JavaScript program to extract all mobile numbers, email addresses, and dates from a paragraph using regular expressions. Display each category separately.

**CODE:**

```
<!DOCTYPE html>
<html>
<body>
<script>
let paragraph = "Contact Ravi at 9876543210 or ravi@gmail.com. Date: 15-08-2026. Call Priya at 9123456780. Email priya@yahoo.com on 20-10-2026.";
let mobiles = paragraph.match(/\b[6-9]\d{9}\b/g) || [];
let emails = paragraph.match(/\b[A-Za-z0-9._%+-]+@[A-Za-z0-9.-]+\.[A-Za-z]{2,}\b/g) || [];
let dates = paragraph.match(/\b\d{2}-\d{2}-\d{4}\b/g) || [];
document.write("<h3>Mobile Numbers</h3>");
document.write(mobiles.join("<br>"));
document.write("<h3>Email Addresses</h3>");
document.write(emails.join("<br>"));
document.write("<h3>Dates</h3>");
document.write(dates.join("<br>"));
</script>
</body>
</html>
```

**OUTPUT:**

